Algebra

CHAPTER 11

Statistics **11.1–11.2**

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CHAPTER 11

Statistics

DPP 11.1

Measures of Central Tendency

Single Correct Answer Type

1. The median of values 10, 14, 11, 9, 8, 12, 6 is

(a) 8 (b) 10
(c) 9 (d) 11

2. Find the mode of the following data: 1, 2, 5, 3, 2, 3, 0, 5, 2

(a) 3 (b) 1
(c) 5 (d) 2

3. If values $x_1, x_2, x_3, \ldots, x_{11}$ occur with the frequencies ${}^{10}C_0$, ${}^{10}C_1$, ${}^{10}C_2, \ldots$, ${}^{10}C_{10}$, respectively, then the mode is

(a) x_5 (b) x_6
(c) x_{10} (d) x_{11}

4. The average salary of 9 men is Rs. x. After another man, M_{10} joins the group, the average salary increases by 5%. Further, another man, M_{11} joins the group, and the average returns to the old level of Rs. x. Then which of the following is true?

(a) The salary of M_{10} is $1.7x$.
(b) The salary of M_{11} is $0.3x$.
(c) The sum of salaries of M_{10} and M_{11} is $2x$.
(d) The average salary of M_{10} and M_{11} is $2x$.

5. The mean of 21 observations (all different) is 40. If each observation that is greater than the median is increased by 21, then the mean of all observations will become

(a) 45 (b) 50
(c) 55 (d) 60

6. A variable takes the values 0, 1, 2, ..., n with frequencies proportional to the binomial coefficients ${}^nC_0$, ${}^nC_1$, ${}^nC_2, \ldots$, ${}^nC_n$, respectively. If the mean of the distribution is 20, then the value of n is

(a) 25 (b) 30
(c) 40 (d) 50

7. In a moderately asymmetric distribution, the difference between mean and mode is 39. The difference between mean and median is

(a) 12 (b) 13
(c) 15 (d) 18

Answer Key

Single Correct Answer Type

1. (b) **2.** (d) **3.** (b) **4.** (c) **5.** (b) **6.** (c) **7.** (b)

DPP 11.2
Measures of Dispersion

Single Correct Answer Type

1. If the algebraic sum of deviations of 30 observations from 40 is 30, then the greatest integer less than or equal to the mean of the observations is

(a) 40 (b) 41
(c) 42 (d) 43

2. The mean deviation from the mean of the observations a, $a + d$, $a + 2d$, ..., $a + 2nd$ is

(a) $\frac{n(n+1)d^2}{3}$ (b) $\frac{n(n+1)}{2}d^2$
(c) $a + \frac{n(n+1)d^2}{2}$ (d) $\frac{n(n+1)|d|}{(2n+1)}$

3. The mean deviation of an ungrouped data is 50. If each observation is decreased by 10%, then the new mean deviation will be

(a) 40 (b) 45
(c) 50 (d) 55

4. The mean deviation about median of observations 1, 2, 3, 4, …, 50 is

(a) 10 (b) 12
(c) 12.5 (d) 20

5. The mean of a distribution is 10 if coefficient of variation is 35%. Then the standard deviation of the distribution is

(a) 3.5 (b) 2.5
(c) 25 (d) 35

6. The standard deviation of the set of observations, 11, 12, 13, 14, 15, is

(a) 1.1 (b) 1.2
(c) 1.3 (d) $\sqrt{2}$

7. Consider the following data:

x_i (observation):	0	1	2	3	4
f_i (frequency):	1	9	7	5	3

The variance of this data is

(a) 1 (b) 1.1
(c) 1.2 (d) 1.3

8. The mean of two samples of sizes 200 and 300 are found to be 5 and 6, respectively. Their standard deviations are 3 and 4, respectively. Then the variance of the combined sample is

(a) 11.76 (b) 12.56
(c) 13.44 (d) 14.24

9. The mean and variance of 10 numbers were calculated as 10 and 5, respectively. It was subsequently found that one of the numbers was misread as 10 instead of 15. The actual variance of numbers is

(a) 5 (b) 6
(c) 7 (d) 7.25

Numerical Value Type

10. The mean and variance of 7 observations are 8 and 16, respectively. If 5 of the observations are 2, 4, 10, 12 and 14, then the difference of remaining two observations is

11. If $n = 10, \bar{x} = 10$ and $\Sigma x^2 = 2000$, then the standard deviation is

12. If $\sum_{i=1}^{9}(x_i - 10) = 9$ and $\sum_{i=1}^{9}(x_i - 10)^2 = 45$, then S.D. of $x_1, x_2, \ldots, x_9$ is

Answer Key

Single Correct Answer Type

1. (b) **2.** (d) **3.** (b) **4.** (c) **5.** (a) **6.** (d) **7.** (c) **8.** (c) **9.** (d)

Numerical Value Type

10. (2) **11.** (10) **12.** (2)

CHAPTER 11

DPP 11.1

Single Correct Answer Type

1. (b) Observations are: 6, 8, 9, 10, 11, 12, 14

$\therefore$ Median = 10

2. (d) Number 2 repeats the maximum number of times.

3. (b) Frequency of $x_6 = {}^{10}C_5$; This is the maximum value.

4. (c) $\dfrac{s_1 + s_2 + ... + s_9}{9} = x$ … (1)

$\dfrac{(s_1 + s_2 + ... + s_9) + s_{10}}{10} = x + \dfrac{1}{20}x$ … (2)

From equations (1) and (2),

$s_{10} = 1.5x$

Now, $\dfrac{(s_1 + s_2 + ... + s_9) + s_{10} + s_{11}}{11} = x$ … (3)

From equations (1), (2) and (3)

$s_{11} = 0.5x$

5. (b) Mean of 21 observations, $\bar{x} = 40$

So $\Sigma x = 21 \times 40 = 840$

If each of the observations greater than the median is increased by 21, then each of the ten observations will increase by 21.

$\therefore$ Sum of all observations = $840 + 10 \times (21) = 1050$

$\therefore$ New mean = $\dfrac{1050}{21} = 50$

6.(c) $\bar{x} = \dfrac{0 \cdot {}^nC_0 + 1 \cdot {}^nC_1 + 2 \cdot {}^nC_2 + ... + n \cdot {}^nC_n}{{}^nC_0 + {}^nC_1 + {}^nC_2 + ... + {}^nC_n} = \dfrac{n \cdot 2^{n-1}}{2^n} = \dfrac{n}{2}$

$\Rightarrow \dfrac{n}{2} = 20$ (given)

$\therefore n = 40$

7. (b) Mean – Mode = 39

We know that

Mode = 3(Median) – 2(Mean)

$\Rightarrow$ Mean – 39 = 3(Median) – 2(Mean)

$\Rightarrow$ Mean – Median = 13

DPP 11.2

Single Correct Answer Type

1. (b) $\sum_{i=1}^{30}(x_i - 40) = 30$

$\Rightarrow \sum_{i=1}^{30} x_i - 40 \times 30 = 40$

$\Rightarrow \sum_{i=1}^{30} x_i = 1200 + 40 = 1240$

So, the mean = $\dfrac{1240}{30} = 41.33$

2. (d) $\bar{x} = \dfrac{1}{2n+1}[a + (a+d) + \ldots + (a + 2nd)]$

$= \dfrac{1}{2n+1}\dfrac{2n+1}{2}[a + a + 2nd] = a + nd$

$\therefore$ M.D. = $\dfrac{1}{2n+1} 2|d|(1 + 2 + ... + n) = \dfrac{n(n+1)|d|}{(2n+1)}$

3. (b) Let $x_1, x_2, x_3, \ldots, x_n$ be the original observations.

Then the new observations will be:

$(0.9)x_1, (0.9)x_2, (0.9)x_3, \ldots, (0.9)\, x_n$

$\therefore$ New mean = $0.9\bar{x}$

$\therefore$ New mean deviation = $\dfrac{1}{n}\sum |0.9x_i - 0.9\bar{x}|$

$= 0.9\,\dfrac{1}{n}\sum |x_i - \bar{x}|$

$= 0.9 \times 50 = 45$

4. (c) $n = 50$

Median = $\dfrac{25^{th} \text{ value} + 26^{st} \text{ value}}{2} = \dfrac{25 + 26}{2} = 25.5$

M.D. (about median) = $\dfrac{\sum_{i=1}^{50}|x_i - 25.5|}{50}$

$= \dfrac{24.5 + 23.5 + ... + 0.5 + 0.5 + ... + 24.5}{50}$

$= \dfrac{1 + 3 + 5 + ... + 49}{50} = \dfrac{625}{50} = 12.5$

5. (a) Coefficient of variation = $0.35 = \dfrac{\sigma}{\bar{x}}$

$\therefore$ S.D. = $\sigma = 0.35 \times 10 = 3.5$

6. (d) Mean = $\dfrac{11 + 12 + 13 + 14 + 15}{5} = 13$

$\therefore$ Variance = $\sigma^2 = \dfrac{(13-11)^2 + (13-12)^2 + (13-13)^2 + (13-14)^2 + (13-15)^2}{5} = 2$

$\therefore$ S.D. = $\sqrt{\sigma^2} = \sqrt{2}$

7. (c) $\sigma^2 = \left[\dfrac{\sum f_i x_i^2}{\sum f_i} - \left(\dfrac{\sum f_i x_i}{\sum f_i}\right)^2\right] = \left[\dfrac{130}{25} - \left(\dfrac{50}{25}\right)^2\right] = 1.2$

8. (c) We have $\frac{\Sigma x_i}{200} = 5$ and $\frac{\Sigma y_i}{300} = 6$

$\Rightarrow \quad \Sigma x_i = 1000$ and $\Sigma y_i = 1800$

Given that $\sigma_x = 3$ and $\sigma_y = 4$

$$\Rightarrow \quad \frac{\Sigma x_i^2}{200} - 5^2 = 9 \text{ and } \frac{\Sigma y_i^2}{300} - 6^2 = 16$$

$\Rightarrow \quad \Sigma x_i^2 = 6800$ and $\Sigma y_i^2 = 15600$

$$\therefore \quad \sigma = \frac{\Sigma(x_i^2 + y_i^2)}{500} - \left(\frac{\Sigma x_i + \Sigma y_i}{500}\right)^2$$

$$= \frac{6800 + 15600}{500} - \left(\frac{1000 + 1800}{500}\right)^2$$

$= 44.8 - 31.36 = 13.44$

9. (d) Let x_{10} (= 10) be the misread value of $(x_{10})_{\text{actual}}$ (= 15).

Given $\sigma^2 = 5$ and $\bar{x} = 10$

$$\Rightarrow \quad \sum_{i=1}^{9} x_i = 100 - 10 = 90 = 10\bar{x} - 10$$

$$\text{Also, } \sigma^2 = \frac{\sum_{i=1}^{9} x_i^2 + x_{10}^2}{10} - (\bar{x})^2$$

$$\Rightarrow \quad \sum_{i=1}^{9} x_i^2 = -50 + 10(\bar{x})^2 \qquad \ldots (1)$$

$$\Rightarrow \quad (\sigma^2)_{\text{actual}} = \frac{\sum_{i=1}^{9} x_i^2 + (x_{10})_{\text{actual}}^2}{10} - [(\bar{x})_{\text{actual}}]^2$$

$$= \frac{-50 + 10(\bar{x})^2 + 225}{10} - \left(\frac{10(\bar{x}) - 10 + 15}{10}\right)^2$$

$$= (\bar{x})^2 + 17.5 - \left(\bar{x} + \frac{1}{2}\right)^2$$

$= 17.5 - \bar{x} - 0.25 = 7.25$

Numerical Value Type

10. (2) Let the two observations be x and y.

$$\therefore \quad \frac{x + y + 2 + 4 + 10 + 12 + 14}{7} = 8$$

$\Rightarrow \quad x + y = 14 \qquad \ldots (1)$

$$\text{and} \quad \frac{x^2 + y^2 + 4 + 16 + 100 + 144 + 196}{7} - \frac{(x + y + 42)^2}{49} = 16$$

$\Rightarrow \quad x^2 + y^2 = 560 - 460 = 100 \qquad \ldots (2)$

Solving (1) and (2), we get $x = 6, y = 8$ or $x = 8, y = 6$.

11. (10) $\sigma^2 = \frac{\Sigma x^2}{n} - (\bar{x})^2$

$$\therefore \quad \sigma^2 = \frac{2000}{10} - 10^2 = 100$$

$\Rightarrow \quad \text{S.D.} = \sqrt{\sigma^2} = 10$

12. (2) S.D.(x_1) = S.D.$(x_i - 10)$

$$= \sqrt{\frac{\sum (x_i - 10)^2}{n} - \left(\frac{\sum (x_i - 10)}{n}\right)^2} = \sqrt{\frac{45}{9} - 1} = 2$$

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